

**5. Ensures computer system and network reliability whether provided directly by the institution or through contractual arrangements.**

The Technical Infrastructure Plan addresses computer system and network reliability, including power and data backups. The plan details cloud services, network agreements, and filtering.

**6. Is evaluated annually (and revised as necessary).**

The IT Committee annually evaluates the Technical Infrastructure Plan and revises it as necessary. The annual evaluation is intended to ensure the plan's purpose is met and the associated resources and infrastructure are functioning as intended. Changes to the plan are approved by the President's Cabinet, which consists of administration, faculty, and staff.

**7. Ensures basic information about the plan is available to the administration, faculty, and staff. [See Section VII. Definitions – Plan]**

The Technical Infrastructure Plan is available on Southwest Technical College's website.

**8. The technology used by the institution to deliver services and, if applicable, program content to students meets the needs of the students without creating barriers to student support or learning.**

Instructional and non-instructional equipment and supplies are used in accordance with manufacturer recommendations and standard practices.

As part of the regular schedule, the Information Technology department performs preventative maintenance and upgrades to the physical resources and technical infrastructure. In the case of highly technical maintenance, repairs, replacement, or disposal of non-instructional equipment and supplies, the work may be contracted with a qualified company.

In the case of specific program equipment (i.e., instructional equipment), the program director or faculty are the experts. In some cases, maintenance or repairs can be performed internally, while other situations may require outsourcing to technical experts. Each program has an equipment budget that is controlled by the program director. During the budgeting process, the program can make requests for the replacement, disposal, and/or upgrade of equipment.

Each department and program budgets for and purchases the supplies that are necessary for its operations. The Director of Information Technology develops an annual budget for IT support function supplies.

**9. Procedures for preserving and protecting student coursework, testing, and records are provided by measures that ensure their preservation and protection.**

Utilizing the SCALE (H3C) virtual environment, all mission-critical servers operate as a virtual environment. A secondary backup array is in one of the college's secondary campus locations, where a snapshot is taken and stored every 15 minutes. In the event of a server or

array failure, critical servers can be brought online using either the primary array or directly on the secondary array. Having this capability reduces the amount of time critical services and data would be unavailable. As a further precaution, full backups are made on a removable drive each night (Monday through Friday). The removable drives can store 30 days of restorable copies. Backups are tested in a sandbox environment to ensure restorability.

## **Challenges and Solutions**

Southwest Technical College's IT department faces a range of challenges, from maintaining an aging infrastructure to adapting to rapidly evolving technologies. Balancing security and accessibility is a constant concern, as we must protect sensitive data while ensuring seamless access for students, faculty, and staff across multiple devices and platforms. Budget constraints often limit our ability to invest in necessary upgrades or staff training, even as the demand for reliable technology grows. Additionally, the IT department must support diverse needs, such as hybrid learning environments, research computing, and campus-wide communication systems, all while addressing cybersecurity threats and meeting compliance requirements. These challenges require strategic planning, cross-department collaboration, and innovative problem-solving to meet the demands of a modern campus.

## **Summary**

Southwest Technical College's program offerings for students are diverse, requiring a significant investment in equipment, technology, lab space, classrooms, testing centers, offices, and general gathering areas. Oversight of these areas is the primary responsibility of the Director of Facilities and Director of Information Technology. The Facilities and IT teams strive to maintain an adequate, safe, and clean environment essential for student learning.